
Two Readouts of a Traffic GNN, and Why Neither Supports a Claim About the Road Network

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Abstract

1 We ask whether two standard post-hoc readouts of a spatio-temporal GNN — a
2 mask-based explainer and the model’s own learned adjacency — yield checkable
3 structure about the road network they were trained on. Neither does. What makes
4 that null worth reporting rather than merely asserting is the instrumentation: three
5 independent stability checks, an external geometric baseline that is independent
6 of the model, a second explainer family, and a pre-registered reliability proxy
7 that failed its own criteria. The same geometric test that finds nothing for the
8 explainer recovers road structure at 0.76–0.81 precision for a distance baseline,
9 so the instrument demonstrably detects real structure when it is present. We also
10 report one apparent effect that a sample-size robustness check did not reproduce,
11 and withdraw it.

12 1 Introduction

13 Interpretability becomes discovery only when a readout says something about the world that can be
14 checked independently of the model that produced it. We take a spatio-temporal GNN trained to
15 forecast traffic speed on the METR-LA sensor network and ask directly: does reading its internals
16 tell us anything testable about the road network? Two readouts are available and both are standard —
17 a GNNExplainer-style soft mask over input sensors, and the adaptive adjacency the model learns for
18 itself — and we find that neither supports such a claim. The result is a null, so the instrumentation
19 carries the weight: we measure stability three independent ways, compare against an *external* check
20 that does not involve the model at all (physical road distance between sensors, taken from the road-
21 distance table the dataset ships), cross-check against a second explainer family, and pre-register a
22 cheap reliability proxy that then fails. That external baseline is what makes the null interpretable rather
23 than uninformative: applied to a nearest-by-road rule it recovers the road graph at high precision, so a
24 near-zero score for the explainer reflects the explainer, not a weak test.

25 2 Setup

26 The forecaster is a Graph WaveNet-style ST-GNN (Wu et al., 2019) on METR-LA (207 loop sensors).
27 Its spatial support mixes a *given* adjacency with a *learned* one as $A_{\text{final}} = \alpha A_{\text{phys}} + (1 - \alpha) A_{\text{sem}}$.
28 The given adjacency is a thresholded Gaussian kernel on directed road distance: with $\sigma = 2584.5$ m
29 and threshold $\kappa = 0.1$ the effective cutoff is 3921.7 m, giving 1,515 directed edges over 207 nodes
30 (density 0.0355). “Off-adjacency” below therefore means precisely “more than about 3.9 km apart by
31 road,” not a vague notion of distance. The learned component is $A_{\text{sem}} = \text{softmax}(\text{relu}(EE^T))$ with
32 E a learned [207, 10] embedding initialised from $\mathcal{N}(0, 0.01^2)$ — not seeded from the kernel — and
33 on the checkpoint we analyse the model puts weight 0.7032 on the learned graph against 0.2968 on
34 the given one. The first readout is a GNNExplainer-style (Ying et al., 2019) node mask, optimised per

(target, window) and read as a top- k ($k=8$) source list; the aggregate over windows is an influence matrix W . Unless stated otherwise every number below comes from one run of $207 \times 24 = 4,968$ mask solves on CPU, and from the same epoch-54 checkpoint, so the two readouts describe one model.¹

3 Result 1: the mask is close to flat

Covering 80% of a target’s off-self importance mass takes on average 146.6 ± 4.7 sources in the congested regime and 154.0 ± 2.9 in free flow, out of 206 available. A readout that needs roughly three quarters of the network to account for four fifths of its mass is not selecting a small causal set, and the narrow standard deviations show this is a property of the method rather than of a few odd targets. The learned adjacency is no better: its rows need 142.1 of 206 sources for the same 80%, it is structurally dense (every entry positive), and each row spreads over 107 of 207 nodes — 52% of uniform. The model’s spatial receptive field is effectively the whole graph in one hop.

4 Result 2: instability, measured three ways

Flat readouts should be unstable, and they are. Table 1 collects the evidence under perturbations that have nothing to do with one another.

Table 1: Stability of the two readouts. Every entry is agreement between two computations that should have given the same answer. “Pooled” is one Jaccard over the union of all targets’ top-8 source–target pairs across the full node set, with no adjacency or reachability restriction; “per-target mean” averages each target’s own Jaccard and is a *different* statistic, hence the spread. Chance references are two independent uniform draws under the same construction.

Perturbation	Statistic	Value	Chance ref.
<i>Mask explainer</i>			
Window resample, free-flow (200 targets)	pooled top-8 Jaccard	0.2256	0.0165
Window resample, congested (28 targets)	pooled top-8 Jaccard	0.1399	0.0182
Window resample, $\lambda/3\lambda/10\lambda$ (40)	per-target mean	0.2323 / 0.2385 / 0.2287	—
Optimiser seed, $\lambda/3\lambda/10\lambda$ (10)	per-target mean	0.3480 / 0.3149 / 0.3015	—
CPU vs MPS, same seed (3 cases)	top-8 Jaccard	0.3991	—
CPU vs MPS, same seed (3 cases)	Pearson, magnitudes	0.9193	—
<i>Learned adjacency</i>			
Epoch 34 vs 54, same run	top-8 Jaccard	0.2163	0.0163
Against a separate training run	top-8 Jaccard	0.1532 / 0.1589	0.0163
<i>Between the two readouts</i>			
W vs A_{sem} , free-flow / congested	per-target top-8 J	0.0873 / 0.1743	—

Three things are worth drawing out. First, raising the sparsity penalty tenfold does not concentrate the mask — per-target agreement is flat across λ , at 0.2323 ± 0.1461 , 0.2385 ± 0.1739 and 0.2287 ± 0.1831 — so the instability is not a tuning artefact that a stronger prior would remove. Second, the CPU-versus-MPS row is the starkest: the same target, the same window and the same seed agree on importance *magnitudes* at Pearson 0.9193 (Spearman 0.5778) while agreeing on top-8 *identity* only 0.3991 of the time (0.3576 at top-20). Floating-point associativity alone reorders which sensors a reader would be told mattered. Third, the learned adjacency preserves *ranks* across training states far better than it preserves top- k identity (per-target Spearman 0.7795 between epochs against a top-8 Jaccard of 0.2163), which is the same dissociation seen in the mask: what is stable is not what gets reported. The two readouts also largely disagree with each other, with 84 of 200 free-flow targets sharing *nothing* in their top 8 (17 of 28 when congested) and per-target Spearman of $+0.0618$ and -0.115 .

¹Explainer solves: run 20260824T235016Z__influence_graph_solves. Learned adjacency: run 20260825T093132Z__learned_semantic_graph, checkpoint metr_la_best.pt, sha 046b2a4a, epoch 54.

5 Result 3: the external geometric check

The checkable question is whether either readout recovers road geometry. We score each method’s top-8 edges against the given kernel adjacency on the same targets (Table 2).

Table 2: Precision against the kernel adjacency, same targets and same $k=8$ for every row. The nearest-by-road baseline is the calibration: it must score well, since the adjacency is a road-distance threshold, and the fact that it does is what shows the test detects road structure when a method encodes it.

Method	Free-flow	Congested
Nearest by road distance (baseline)	0.7557	0.8088
Mask explainer, top-8	0.2863	0.1397
Degree-matched random	0.0344	0.0551
Uniform random	0.0331	0.0368

The explainer sits above both random floors, so its mask is not noise — but it recovers a small fraction of what a plain distance rule recovers on the identical test. **That gap is the paper’s calibration point:** the instrument is demonstrably able to detect road structure, so a low score for the explainer reflects the explainer rather than a weak test. The learned adjacency does worse still. Its top-8 overlaps a nearest-by-road rule at 0.0644, and it places 0.0445 of its mass on adjacency edges against a 0.0355 chance rate — a lift of 1.252, close to none. Nor does the explainer favour long-range structure, which would have been a different result worth reporting: beyond eight chained kernel radii it is *depleted*, at $0.570\times$ the base rate in free flow and $0.923\times$ when congested.

6 Result 4: a second explainer family disagrees, invisibly

On a 28-scenario subset where SHAP (KernelExplainer) was run against the same model, the two methods’ top- k sets overlap at a mean Jaccard of 0.0221. Two principled methods for reading one model agree on about 2% of what they nominate. Yet a downstream consumer cannot tell: a deterministic citation checker scoring an LLM’s narration of each returns *identical* precision 0.9821 and hallucination 0.0179 for both.² Whatever disagreement exists between explainer families is not visible to anything built on top of them. *Checkpoint caveat:* this comparison ran on the epoch-34 checkpoint, not the epoch-54 one used elsewhere in this paper. Both of its arms use that same checkpoint, so the 0.0221 overlap is an internally consistent within-model comparison; but it is a different training state from Results 1–3 and we do not pool it with them. The summary file records no checkpoint, so we established this by re-solving cached explanations on both candidates: they reproduce exactly on epoch 34 (top-8 Jaccard 1.000) and not on epoch 54 (0.067 and 0.000).

7 Result 5: an apparent effect, withdrawn

We report one result that did not survive its own robustness check, because the check is the point.

Restricting to explainer edges beyond eight chained kernel radii and asking which survive both split halves, 52 of 423 free-flow edges survive. Those survivors sit in the learned adjacency’s own top-8 at a rate of 0.1538, against 0.0404 for edges surviving only one half — an apparent shift of roughly four-fold, which would have been the most interesting thing here: two independent readouts agreeing about long-range structure that the road graph does not contain.

Recomputing on the first 64 targets of the same seeded shuffle — an unbiased prefix, so this isolates sample size rather than geography — the shift did not reproduce: survivors sit in the top-8 at 0.0714 against 0.0381 for one-half-only, on 14 survivors instead of 52. The congested regime shows 0.0 at both sample sizes. We therefore withdraw the effect. Two readings are consistent with the data — the prefix is underpowered at 14 survivors, or the full-sample shift is not robust — and neither is established, so the honest statement is that the effect has not been shown to replicate and the four-fold figure should not be quoted on its own. We flag the direction explicitly because it is the opposite of the more familiar failure: here the apparent effect is at the *larger* sample and the subsample is what

²shap_comparison_summary.json; $n=28$.

100 fails to reproduce it, which makes it ambiguous rather than cleanly negative. The next section is the
101 cleanly negative case.

102 **8 Result 6: a pre-registered reliability proxy, falsified**

103 If flatness drives instability then mask entropy should predict it, and entropy costs one solve where
104 measuring stability costs two. We fixed the criteria before running: Spearman $\rho < -0.30$ against
105 the split-half Jaccard *and* a bootstrap CI excluding zero, plus a threshold chosen on the sweep alone
106 flagging all 12 committed explanations. Pooled over 120 (target, setting) pairs, $\rho = -0.2008$ with
107 95% CI $[-0.4229, +0.0452]$ for the entropy of the window-averaged mask, and $\rho = -0.1732$, CI
108 $[-0.3938, +0.0513]$ for the mean per-window entropy — the latter being the only form obtainable
109 from a single solve, and so the only one that would have made the proxy cheap. Both intervals include
110 zero and the threshold flagged 3 of 11 distinct committed explanations. Both criteria are falsified. A
111 single-setting version at $n=40$ did appear to pass, at $\rho = -0.3412$ with 95% CI $[-0.5999, -0.0241]$,
112 meeting both criteria; it did not survive the full pool.³ That is the expected fate of an underpowered
113 positive, and the reason the criteria were fixed in advance.

114 **9 Discussion**

115 Three things would make either readout usable for a claim about the road network: stability under
116 perturbations that should not matter, agreement across explainer families, and agreement with a check
117 external to the model. None holds here. Top- k identity moves under window resampling, optimiser
118 seed, and floating-point associativity alike; two explainer families overlap at 2%; and the external
119 geometric test, calibrated by a baseline that passes it comfortably, finds little.

120 The general lesson is about consumption, not about this model. An interpretability output that is
121 handed downstream — to an LLM that narrates it, to an automatic checker that scores the narration,
122 to a human reading a ranked list — carries its instability with it, and nothing downstream surfaces
123 that. The SHAP comparison is the sharpest version: two readouts that disagree almost completely
124 produce byte-identical downstream scores. Any pipeline that turns an attribution into a claim should
125 therefore carry its own stability estimate alongside the attribution, or the claim inherits an error bar
126 nobody has measured.

127 **10 Responsible use statement**

128 This work concerns interpretability readouts of a traffic-forecasting model, in a research setting, on a
129 public sensor dataset with no personal data. Our results should not be read as a safety claim about the
130 model or the explainer for their original purpose: we did not evaluate forecasting accuracy here, and
131 the absence of a validated discovery does not imply the forecaster is unfit for forecasting or that the
132 explainer is unfit for the debugging and sanity-checking uses it is normally put to. The scoped claim
133 is narrower and concerns downstream use: these particular readouts should not be treated as evidence
134 about real road structure — for instance to justify a traffic-management intervention at a sensor
135 named in a top- k list — without the stability and external checks demonstrated here. Presenting an
136 unstable attribution to an operator as though it were a determined fact is the specific harm this paper
137 argues against.

138 **11 Limitations**

139 **(a)** One dataset. Everything here is METR-LA; we make no cross-city claim, and whether these
140 nulls hold on another sensor network is untested. **(b)** The external baseline inherits a definition.
141 “Geometry” here means the dataset’s directed road-distance table thresholded at 3921.7 m, which is
142 one sensible notion and not the only one; moreover only 27.1% of ordered sensor pairs have a road
143 distance recorded at all (24.4% of non-adjacent pairs), so the baseline is evaluated on the covered

³Recomputed into its own run directory 20260827T110720Z__entropy_gate_single_setting from the Stage-1 masks, using the same functions as the pooled analysis, because the original inline value had been overwritten by the pooled result.

subset. (c) Some checks are small: the seed-repeat uses 10 targets, the cross-device comparison 3 cases, the SHAP comparison 28 scenarios, and the congested split-half only 28 targets. These are indicative. The central claims rest on the full-sample split-half (200 free-flow targets) and the geometry comparison, both at $n=207$ solved targets. (d) A single explainer family plus one SHAP cross-check; other attribution methods might behave differently, which is exactly what Result 4 suggests is worth testing.

12 Related work

Evaluating GNN explainers against ground truth is itself contested. Sanchez-Lengeling et al. (2020) benchmark attribution against computable ground-truth attributions on synthetic graphs and Agarwal et al. (2023) supply generators and a library for the same purpose, while Faber et al. (2021) show that comparing explanations to ground truth is unsound when the trained GNN does not use the ground-truth edges, separating the responsible, causing and explanatory motifs. Our setting has no ground-truth attribution at all, which is why we substitute an external *geometric* check and calibrate it with a baseline that passes.

The second readout we examine is the adaptive adjacency now standard in spatio-temporal forecasting: Graph WaveNet (Wu et al., 2019) introduced a learned self-adaptive matrix, MTGNN (Wu et al., 2020) learns a unidirectional graph from node embeddings, and AGCRN (Bai et al., 2020) learns node-specific parameters and infers inter-series dependencies without a predefined graph. These are often described as capturing hidden spatial dependency, which invites reading the learned matrix as a discovered graph. Our measurements argue against that reading for this model: the learned matrix is structurally dense, spreads 52% of uniform across each row, aligns with road adjacency at a lift of 1.252, and its top-8 changes substantially between training states.

References

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A Claim-to-source map

Every number in the main text traces to one of the following. Run directories are under `evaluation/results/`; IG abbreviates `20260824T235016Z__influence_graph_solves`, LG abbreviates `20260825T093132Z__learned_semantic_graph`, and GWI abbreviates `20260825T201433Z__grounding_without_information`.

Claim	Value	Source
Kernel adjacency construction	σ 2584.5 m, κ 0.1, cutoff 3921.7 m, 1515 edges, density 0.0355	IG metrics.json, geometry
α split, learned vs given	0.7032 / 0.2968	LG learned_graph.json, per_checkpoint.metr_la_best.pt.alpha
Mask flatness, 80% mass	146.6 $\pm$ 4.7 congested; 154.0 $\pm$ 2.9 free-flow, of 206	IG metrics.json, per_regime*.sources_for_mass_frac
Learned graph flatness	142.1 of 206; 107 of 207 per row (52% of uniform)	LG, sources_for_mass_frac, density
Split-half J, <i>pooled</i>	0.2256 (200 targets) / 0.1399 (28); ref 0.0165 / 0.0182	IG metrics.json, per_regime*.stability.jaccard_all_edges
Split-half J, <i>per-target mean</i> , 3 λ	0.2323 / 0.2385 / 0.2287	GWl part4_metrics.json, settings[]
Seed-repeat J, 3 λ	0.3480 / 0.3149 / 0.3015 ($n=10$)	GWl part4_metrics.json
CPU vs MPS	Pearson 0.9193; top-8 J 0.3991; Spearman 0.5778; top-20 J 0.3576	20260824T235643Z__device_divergence, device_divergence.json
Cross-checkpoint J, learned graph	0.2163 / 0.1532 / 0.1589; ref 0.0163	LG, cross_checkpoint_stability
W vs A_{sem} agreement	J 0.0873 (84 zero) / 0.1743 (17 zero); Spearman +0.0618 / -0.115	IG metrics.json, per_regime*.vs_learned_semantic_graph
186 Geometry precision, explainer	0.2863 free-flow / 0.1397 congested	IG metrics.json, per_regime*.edges.top_k.precision
Geometry precision, baselines	nearest-road 0.7557 / 0.8088; uniform 0.0331 / 0.0368; degree 0.0344 / 0.0551	IG metrics.json, per_regime*.baselines
Learned graph vs geometry	overlap 0.0644; mass 0.0445 vs 0.0355 chance, lift 1.252	LG, overlap_with_nearest_road, road_alignment_semantic
Beyond-8-radii enrichment	0.57 $\times$ free-flow; 0.923 $\times$ congested	IG metrics.json, per_regime*.beyond_k
SHAP overlap and downstream scores (epoch 34)	J 0.0221; precision 0.9821; halluc 0.0179; $n=28$	shap_comparison_summary.json; epoch established by re-solve, see §6
Survivors, full sample	52 of 423; 0.1538 vs 0.0404; survival 0.1229	IG metrics.json, per_regime.free_flow.beyond_k_survivors
Survivors, 64-target prefix	14 survivors; 0.0714 vs 0.0381; congested 0.0 both	IG metrics.json, survivor_sample_size_sensitivity
Entropy proxy ρ and CIs	-0.2008 [-0.4229, +0.0452]; -0.1732 [-0.3938, +0.0513]; 3/11	GWl verdicts.json, P4
Entropy proxy at $n=40$	ρ -0.3412, CI [-0.5999, -0.0241]	20260827T110720Z__entropy_gate_single_setting, entropy_gate_single_setting.json
Road-distance coverage	27.1% of pairs; 24.4% of non-adjacent	IG metrics.json, geometry